K conversion: column touches to rho steps

Result. Measured with Callgrind on this host, one binary. One rho step costs 6665 Ir at n = 83 and 725 Ir at n = 61. One column of the dense degree-3 closure costs 26,000 to 105,000 Ir on the m = 4 cells, so K (rho steps per column touch) is **4.0 to 15.6** with the n = 83 step (pooled m = 4 median 4.48, range 3.96 to 16.77), and K grows with matrix size: 83 at 17,902 columns (degree 4, N = 26). K is above 1, so the conversion **raises** the priced rows, the opposite of the sign the note guessed (its K is the reciprocal).

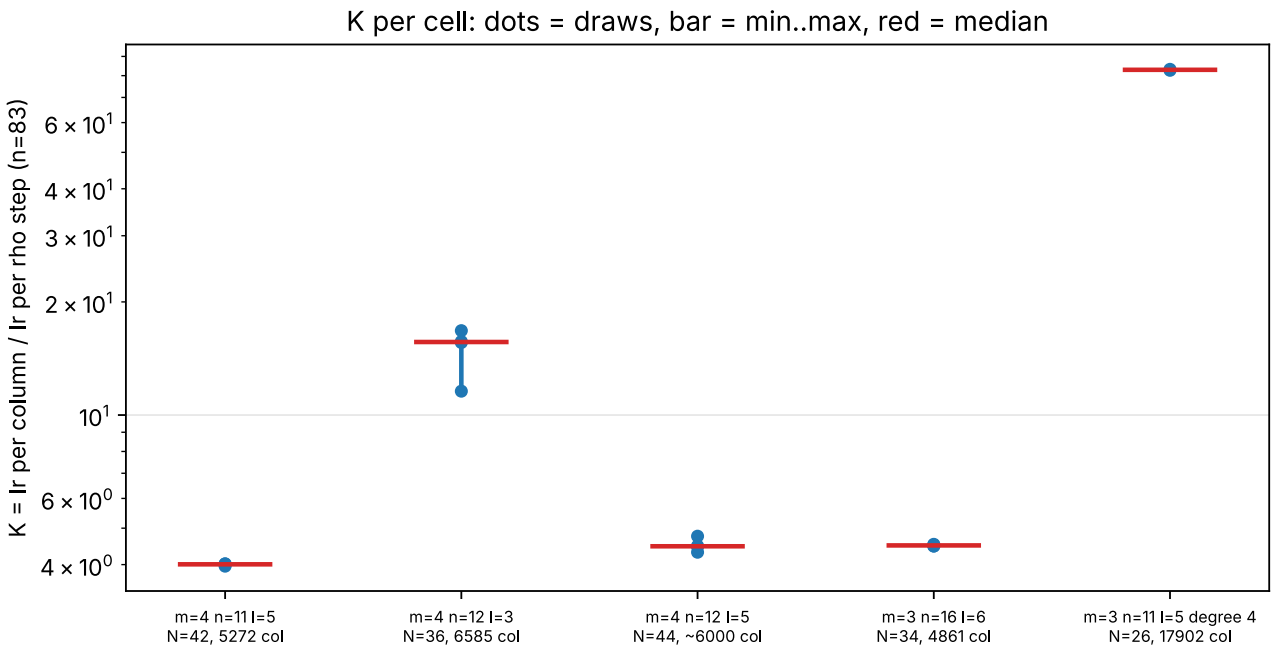


Figure 1: K per cell

Definitions (frozen in PROTOCOL.md before any run)

K = (Ir per column touch) / (Ir per rho step). A column touch is one column of the degree-D Macaulay matrix (the engine's `columns` = C(N, <=D), the note's w = 1 oracle unit). Ir per column = Callgrind Ir inside the dense full-MutantXL closure (including the matrix build) divided by `columns`. A rho step = Ir inside `StrongRhoG::solve` divided by the solver's `walk_steps`. Units are `callgrind.Ir`, simulated user-space instructions; no wall-clock claim.

K per cell (4 draws each; median [min, max])

cell	columns	Ir per column	K, n = 83 step	K, n = 61 step
m4-11-5	5272	26407 to 26786	4.01 [3.96, 4.02]	36.81 [36.40, 36.92]
m4-12-3	6585	77150 to 111783	15.64 [11.57, 16.77]	143.74 [106.34, 154.08]
m4-12-5	6305-6510	28760 to 31728	4.48 [4.31, 4.76]	41.12 [39.64, 43.73]
m3-16-6	4861-4861	29871 to 30186	4.50 [4.48, 4.53]	41.33 [41.17, 41.61]
m3-11-5-deg4	17902	550987 to 554271	82.91 [82.66, 83.16]	761.74 [759.48, 764.01]

Rho step: n = 83 capped walk of 10^6 steps, seeds 1 to 3: 6665.4, 6666.1, 6663.0 Ir/step. n = 61 complete solves, seeds 1 to 3 (all verified): 724.3, 725.5, 725.6 Ir/step. The ecbench record (`research/ecbench_rho_optimized_20261014`) gives 1003.5 Ir/step on another curve, binary and compiler; it is not mixed in.

Controls

- A/A: the same closure draw twice differs by 0.0048%; the same rho seed twice is identical (441330607 Ir both).
- Positive control: Ir inside the collected interval tracks the engine's own `closure_word_ops` across the 16 degree-3 draws (Spearman 0.83).
- Null control: the collected interval is 75 to 79% of total process Ir (the rest is parsing, system build and the plain XL basis).
- Cell m4-12-3 (l = 3, a denser system: 6585 columns, 18 to 29 M word-ops) has K about 3.5 times the other three, which sit within 12% of each other. Within-cell draw spread is 1% to 45%.

Derived conversion of the two ECC2K-130 rows

This is **derived arithmetic**: row + log2 K, on rows the descent-degree note priced and on K measured at N = 26 to 44. It is a change of unit (accounting), not an improvement. The degree question (does the closure stay at degree 4, or 5, up to N near 394) is unchanged and unmeasured; rho stays 2^60.81 and its n = 131 step cost is unmeasured.

row	K used	K	log2 K	converted row	converted gap to rho
degree 4, w=1, ell=33	low(min,n83)	3.96	1.99	2^70.42	+9.61
degree 4, w=1, ell=33	primary(median,n83)	4.48	2.16	2^70.59	+9.78
degree 4, w=1, ell=33	high(max,n83)	16.77	4.07	2^72.50	+11.69
degree 4, w=1, ell=33	n61 median	41.12	5.36	2^73.79	+12.98
degree 4, w=1, ell=33	deg4 growth cell n83	82.91	6.37	2^74.80	+13.99
degree 5, w=1, ell=34	low(min,n83)	3.96	1.99	2^73.12	+12.31
degree 5, w=1, ell=34	primary(median,n83)	4.48	2.16	2^73.29	+12.48
degree 5, w=1, ell=34	high(max,n83)	16.77	4.07	2^75.20	+14.39
degree 5, w=1, ell=34	n61 median	41.12	5.36	2^76.49	+15.68
degree 5, w=1, ell=34	deg4 growth cell n83	82.91	6.37	2^77.50	+16.69

Reading: the old gaps were +7.62 and +10.32 bits. At the pooled median K the converted gaps are +9.78 and +12.48; across the m = 4 degree-3 range +9.61 to +11.69 and +12.31 to +14.39. Because K grows with matrix size and these rows have about 2^49 columns, the rows with K from small cells are **lower bounds** on the oracle side, and the n = 131 step (wider field) would make K smaller by an unmeasured factor. No conversion measured here brings either row below rho.

Limits

- n = 131 rho is not supported by the binary (group order exceeds u128): the denominator is n = 83 (primary) or n = 61.
- Degree 3 cells for the m = 4 numbers; degree 4 only at N = 26 (two draws). Degree 4 at N >= 34 costs too much under Callgrind.
- Symbol deviation from PROTOCOL.md 2.1: `from_system_closed_full` is inlined, so the toggle symbol is the function it calls, `ReducedBasis::from_system_closed_impl`. Same interval.
- Host: shared 4-vCPU Xeon 2.10 GHz VM, no PMU, no isolation tier; Ir is contention-independent. Binary and versions in PROVENANCE.txt. Conductor not installed in this container.
- The “(11,4), (12,4)” cells were read as n = 11 and n = 12 at m = 4.

Files: PROTOCOL.md, raw.jsonl, RESULT.json, run_k.py, analyze.py, make_report.py, PROVENANCE.txt. Next improvement to try: measure K at degree 4 for N >= 34 on a larger host, and an n = 131 wide rho step.